

BTM710 – Workshop #1: Types of Research

Divide into groups. For each of the 2 citations in the References below:

- 1) State the topic in a single sentence

Aleisa et al.	Li et al.
Topic: This paper reviews other peer-reviewed articles to identify and bring focus to the existing and missing privacy solutions for the Internet of Things.	Topic: A new interaction method This paper introduces a new interaction method so that users can trigger shortcuts or actions in a more efficient manner so that they can act more freely.

- 2) As a group, discuss and state the research question(s) (or “knowledge gaps”) that the work described is addressing

Aleisa et al.	Li et al.
Question(s): What are the privacy features that have been proposed and implemented for IoT. Do they satisfy existing privacy principles? What areas of security and privacy remain unexplored and will require greater focus? What needs to be implemented in order to solve privacy concerns for users.	Question(s): To find a more efficient interaction method utilizing more than the N93 interface provides

- 3) List the research methods used to address the questions/gaps

Aleisa et al.	Li et al.
Methods: This paper is a review article that draws information from many other articles related to the topic. They state they reviewed 122 peer reviewed papers.	Methods: They performed their own experiments using willing participants. They also referenced previous studies and other articles related to their topic.

- 4) As a group, discuss and write down the significance of the research in advancing knowledge on the topic and/or technology.

Aleisa et al.	Li et al.
Significance: The paper identifies the current gaps in research regarding privacy with the Internet of	Significance: A way to implement virtual interfaces to utilize the spatial awareness and kinesthetic

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Things, in particular, the assumption that users will be willing to spend time and energy to engage with the privacy features, and overlooks complacency and unwillingness to engage. It also identifies areas that are poorly described in existing literature, and what kind of changes should be made both in the technologies as well as legislation.	memory of the user to allow for more efficient long term usage, improving upon the existing N93 interface.
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- 5) As a group, discuss and write down your thoughts on the level of application of the research to practice (applied, pure, or a combination).

Aleisa et al.	Li et al.
Applicability: I think the paper is a combination of both applied and pure research, though it leans towards the applied research side, as it is conducted with a practical use-case in mind. It is driven by a desire to solve a specific problem, in this case the lack of privacy with the Internet of Things.	Applicability: The research is a combination of applied and pure research where it utilizes previous studies to prove the case of eyes-off interactions while using experiments to show the usefulness of it.

- 6) As a group, discuss and decide where the research fits on the theoretical versus experimental spectrum.

Aleisa et al.	Li et al.
Applicability: This research article fits clearly on the theoretical spectrum, as they do not perform any experiments of their own, but rather collect and cross correlate information from many other articles.	Applicability: The paper outlines both the principles of design surrounding the N93 interface, as well as conducts an experiment that was conducted which involved training participants to use the virtual shelves interface. This makes the research fit strongly to the experimental side of the spectrum.

Submit, on Blackboard as a group (only one person need submit for the group). Be prepared to discuss with the instructor and/or the rest of the class, time permitting.

Remember that this exercise aims to get you to start thinking about research methods, but does not in any way test your knowledge on the subject. You are only being asked to intuitively interpret the research and share your thoughts.

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References

[Aleisa et al] "Privacy of the Internet of Things: a systematic literature review (extended discussion)." *arXiv preprint arXiv:1611.03340* (2016). <https://arxiv.org/pdf/1611.03340>

[Li et al] "Virtual shelves: interactions with orientation aware devices." *Proceedings of the 22nd annual ACM symposium on User interface software and technology*. 2009.
<https://www.dgp.toronto.edu/~dearman/papers/2009-VirtualShelves-UIST.pdf>